Learning Objective 3.3: I can implement an FM receiver in GNU Radio, demodulate a live broadcast FM signal, and decode its RDS data.

Learning Objective 3.4: I can explain and demonstrate the trade-off between bandwidth and noise performance for AM vs. FM.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **FM bandwidth.** Carson's rule estimates the bandwidth of an FM signal with peak deviation Δf and audio bandwidth W as

$$B \approx 2(\Delta f + W)$$

(a) Fill in the table. For the AM row, the station sends the same audio as DSB-AM.

Service	Δf (kHz)	W (kHz)	Bandwidth (kHz)
Broadcast FM, mono	75	15	
Two-way radio, NBFM	5	3	
AM, broadcast audio	–	15	

(b) Open the FM Explorer on the Lesson 7 page and choose the broadcast preset, 75 kHz of deviation with a 15 kHz tone. Read the power inside Carson's band, and explain in words why the signal is wider than $2\Delta f$ = 150 kHz.

2. **The WBFM receiver.** An RTL-SDR records 2.4 MS/s centered on 98.8 MHz, and the station we want is at 99.1 MHz. The receiver is `L07-wbfm-rx.grc`.

   (a) Find the center frequency of the Frequency Xlating FIR Filter, and the sample rate after it decimates by 10.

   shift = ____

   rate = ____

   (b) A classmate decimates by 20 instead. Explain what goes wrong, using the channel filter's ±90 kHz passband and the 57 kHz RDS subcarrier.

   (c) Find the Quadrature Demod gain that makes 75 kHz of deviation produce an output of 1, and the output when the carrier sits 30 kHz above center.

   gain = ____

   output = ____

   (d) Find the decimation from the demodulator to a 48 kHz sound card, and the corner frequency of a 75 μs de-emphasis filter.

   decimation = ____

   corner = ____

   (e) Explain why the flowgraph needs no Throttle, whether it plays a file or receives from the RTL-SDR.

3. **RDS.** RDS sends 1187.5 bits per second on a 57 kHz subcarrier, in groups of 104 bits.

   (a) The RDS branch resamples to 19 kHz. Find the samples per bit, and the time one group takes.

   (b) The station name is eight characters, sent two per group. Radiotext is up to 64 characters, sent four per group. Find the shortest time to receive each, if the station sent nothing else.

   name =

   text =

   (c) The RDS signal is BPSK, a DSB-SC signal. Name the block in `L07-rds-rx.grc` that recovers its carrier, and explain why a recovered carrier 180° off does no harm, given the differential coding.

   (d) A station's audio is clear, but the station name never appears. Explain why RDS needs a stronger signal than the audio does.

4. **AM and FM in the same noise.** `L07-am-vs-fm.grc` sends a 1 kHz tone as AM at $m = 1$ in a 10 kHz channel and as FM in a $2(\Delta f + 5)$ kHz channel, with the same transmitted power and the same noise. It measured these output SNRs in dB:

C/N in 10 kHz (dB)	30	20	15	10
AM	29.5	19.4	14.4	9.2
FM, Δf = 5 kHz	35.5	25.5	20.5	14.2
FM, Δf = 25 kHz	48.5	39.2	26.1	9.3
FM, Δf = 50 kHz	53.2	39.5	17.3	6.4

(a) Find each FM channel's bandwidth, and how many dB more noise it admits than the 10 kHz AM channel.

(b) An FM receiver reaches threshold when its C/N in its own bandwidth falls to about 10 dB. Find the C/N in 10 kHz at which each FM signal reaches threshold, and check each against the table.

(c) A link to a distant receiver runs at 10 dB C/N in 10 kHz. Use the table to choose AM or one of the three FM deviations, and explain the choice in words.

(d) Above threshold, each doubling of the deviation improves FM's output by about 6 dB. Check that rule against the 30 dB column from 5 to 25 kHz.

5. **An AM receiver for ATIS.** An airport broadcasts its ATIS on 119.25 MHz. The RTL-SDR runs at 2.4 MS/s centered on 119.0 MHz, and the receiver is `L07-am-rx.grc`.

   (a) Find the Frequency Xlating FIR Filter's center frequency, and the decimation that takes it straight to a 48 kHz sound card.

   shift = ____

   decimation = ____

   (b) Lesson 6 detected the envelope with Abs, a low-pass filter, and a gain of $\pi/2$. Explain why Complex to Mag replaces Abs and the $\pi/2$ gain.

   (c) Explain why the AM receiver needs an AGC and the FM receiver does not.

Documentation: _____